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A transformer-based few-shot learning pipeline for barley disease detection from field-collected imagery

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ABSTRACT

Artificial intelligence (AI) and deep learning (DL) for plant disease detection are emerging research areas. DL methods generally require a large amount of annotated data for training, which is often costly, time-consuming, and infeasible. This article addresses the data scarcity problem and proposes a few-shot learning (FSL) method for barley plant disease detection. To prepare a dataset, we captured images from outdoor test-bed trials (at two different growth stages of plants across multiple paddocks) under various weather conditions, such as sunny and cloudy. The images are divided into patches and manually labelled into five classes: no-disease, net form net blotch (NFNB) (which is classified into two stages: early and severe), spot form net blotch (SFNB), and scald. We name this as the Barley dataset. We also used the publicly available cassava dataset, which has five classes. The datasets are then applied to the proposed FSL pipeline, which only uses as few as five images for each class in training. We use the Swin transformer as the network backbone. The method with the Swin-B variant as the feature extractor achieved a detection accuracy of 91.80% and 97.93% on the barley disease and cassava datasets, respectively. The result indicates that our FSL model can efficiently perform and classify barley disease with small training data.

1. Introduction

Barley (*Hordeum vulgare L.*) is one of the primary crops worldwide (Rezaei et al., 2024b). However, diseases and pest infestations threaten global crop production. For example, net blotch of barley, caused by the fungal phytopathogen Pyrenophora teres and widespread in many world regions, is one of the most critical threats to barley production, accounting for 10 – 40% yield losses (Alhashel et al., 2023). Net form net blotch (NFNB) and spot form net blotch (SFNB) are two identified barley diseases caused by P. teres f. teres (Ptt) and P. teres f. maculata (Ptm), respectively (Alhashel et al., 2023). These two forms of net blotch are genetically distinct and can be separated by symptoms (Alhashel et al., 2023). NFNB symptoms include a net-like appearance of elongated dark-brown lesions (Alhashel et al., 2023). In comparison, SFNB signs are elliptical dark-brown lesions (Alhashel et al., 2023). Another significant barley disease is barley scald, caused by Rhyncosporium secalis, especially in temperate and wet regions (Ababa et al., 2023). Reportedly, barley scald can cause severe yield losses where barley is cultivated without crop rotation (Ababa et al., 2023). The symptoms are water-soaked grey-green lesions with distinguishable brown margins, and shapes are ovate to irregular (Ababa et al., 2023). Global demand for high-quality barley necessitates advanced management systems to preserve barely production against these diseases. Consequently, the major effort in this study is to distinguish these barley diseases using novel methods.

Given the negative impacts on agricultural products, timely diagnosis of plant diseases is one of the most crucial challenges in crop production to maintain the sustainability of the food supply. Ongoing observation is a conventional way of plant disease identification since morphological characteristics of a plant change when disease pathogens infect it, and particular symptoms appear on different parts of the plant, especially leaves. Colour and shape changes (spots, streaks, yellowing, chlorosis, distortion, cankers, dwarfing, and necrosis) are regular transparent disease signs (Weng et al., 2021). Although visual monitoring,

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endorsed by laboratory examination, is widely used in agriculture, it is arduous and challenging, even in small-scale fields, because it depends on the grower's or advisor's expertise (Sajitha et al., 2024). In addition, laboratory-based methods require complicated sampling, handling, and analytical procedures alongside advanced equipment to reliably identify diseases. As a result, the next generation of plant disease detection methods should address these issues and offer rapid and affordable tools to detect diseases automatically, even when facing deficient data.

Nowadays, sophisticated artificial intelligence (AI) approaches can automate operations and revolutionise agriculture by substituting manual processes. AI applications are vast in agriculture, and it has solved several practical issues, such as pest and object detection (Amrani et al., 2024) and forecasting plant frost (Shammi et al., 2024). In addition, AI techniques have been applied as a non-destructive and non-invasive method of plant disease detection, and computer vision can assist in identifying diseases using visual symptoms (Rezaei et al., 2024a). Data acquisition, pre-processing, feature extraction, and classification are typical steps in plant disease detection. The most common inputs are RGB and hyperspectral images of plant leaves because they are the most accessible parts that can disclose disease signs. Machine learning (ML) developments have addressed some uncomplicated vision-based plant disease detection in agriculture (Atsmon et al., 2024). Moreover, deep learning (DL) techniques have shown promising performance in discriminating patterns and abnormalities from images, which makes DL effective enough in simulating complex models. Well-known architectures, including VGG, DenseNet, ResNet, and Inception, have been implemented to detect plant diseases (Zeng et al., 2024; Dai et al., 2024; Fan et al., 2022).

However, traditional data-driven methods function poorly with little data (Song et al., 2023). Millions of networks' parameters must be trained by large datasets to avoid overfitting, so the performance of DL networks is highly dependent on the training dataset's size (Rezaei et al., 2024a). This remains a significant challenge even with the recent developments since these methods may yield unsatisfactory results if the data is insufficient (Abade et al., 2021). Various scenarios, such as transfer learning and data augmentation techniques, have been investigated to solve this data issue. To avoid overfitting, methods have applied transfer learning that adapts a model, pre-trained on a large source domain, to a target domain (Liu et al., 2024a). For example, knowledge of well-known pre-trained models, including VGG16, VGG19, DenseNet, and InceptionV3, was transferred to apply in disease classification (Fan et al., 2022). A network pre-trained on a large dataset typically outperforms since it cannot be trained on a small target domain (Lin et al., 2022). The issue is that these methods may fail if applied directly to other target domains (Fan et al., 2022). While transfer learning cannot acquire discriminative features in the target domain, limited data is used for fine-tuning (Fan et al., 2022). Given the small target dataset, a self-supervised contrastive learning method with a domain adaptation stage (CLA) was proposed to prevent the performance drop due to this domain shift issue (Zhao et al., 2023). Although the method mitigated the adverse impact of domain shift for disease identification in digital farming, future networks must generalise feature embedding ability to classify diseases with limited samples and quickly adapt to new cases with few examples, in which the performance is independent of the dataset size.

Few-shot learning (FSL), an application of meta-learning, has been designed to classify based on past knowledge and tackle challenges with inadequate data (Hu et al., 2022). The goal of meta-learning, or learning-to-learn, is to enable models to learn, organise and compare objects given past knowledge, like humans, and generalise the knowledge to new unseen objects (Hu et al., 2022). Meta-learning incorporates meta-training, episodic training to learn every episode's task, and meta-testing, which evaluates the trained model (Liu et al., 2023). FSL typically calculates a similarity between the input and provided class examples to categorise objects. While traditional DL networks depend on dataset size, FSL does not rely on dataset size and works given insufficient inputs (Song et al., 2023). Metric-based networks, including Siamese networks, Matching networks, and prototypical networks, are widely employed to tackle the cost of data collection (Vinyals et al., 2016; Chen et al., 2018).

In agriculture, acquiring and annotating extensive datasets of plant disease is expensive and may not be feasible in certain situations, such as when specific diseases are rare. Besides, known diseases may reappear as new incursions with symptom variations. Consequently, FSL is a feasible method to address this challenging issue. FSL can generalise the classification task to new diseases provided with a few training data samples. This is important because plant diseases constantly mutate and may display unseen symptoms. Siamese networks were used to detect plant disease where a pre-trained InceptionV3 was employed as the feature backbone with fine-tuning for disease classification (Argüeso et al., 2020). Although this method outperformed classical transfer learning, it needs a large amount of training data, e.g., at least 100 samples per class, for a reasonable detection accuracy. Nuthalapati and Tunga (2021) suggested adding an embedding feature transformer to provide a self-attention mechanism, which may produce richer representations of each class and better discriminative features (Nuthalapati and Tunga, 2021). Local feature-matching conditional neural adaptive process (LFM-CNAPS) was proposed to reduce the noise influence in FSL (Chen et al., 2021). The proposed network was equipped with a pre-trained ResNet18, an encoder and a FILM layer to extract class features (Chen et al., 2021). Given the small dataset size, they reduced the trainable parameters to avoid overfitting (Chen et al., 2021). A CNN-based shallow network was trained using meta-learning to reduce overfitting caused by the small dataset size (Li and Yang, 2021). They used pre-training, considering that the target domain samples were insufficient (Li and Yang, 2021). To mitigate the disadvantage of dataset size, Li and Chao (2021) developed a semi-supervised FSL method that employed numerous unlabelled samples to improve the performance. These aforementioned methods have focused on datasets captured under laboratory circumstances. Consequently, in-field visual samples may restrict their performance as images collected in uncontrolled outdoor environments can reduce the accuracy of the models (Rezaei et al., 2024b).

Some methods leveraged additional data, such as image description. However, additional data requires more effort than data collection. Such image labelling with additional texts, for example, image captioning, may be impracticable. A generating network based on conditional adversarial autoencoders (CAAE) was proposed by Zhong et al. (2020). Semantic attributes are used as auxiliary data to improve outcomes (Zhong et al., 2020). Another method developed an image-text collaborative representation learning network (ITC-Net), which requires the text modal information of visual inputs as side information (Wang et al., 2021). Cao et al. (2023) used multi-modal input (image-text-label information) for the detection of cucumber disease. Vision-language pre-training (VLP) was implemented to calculate the similarity of samples in the image-text-label space and extract semantic relations between different modalities (Cao et al., 2023). Some methods leverage the frequency domain information to the spatial domain. However, they require additional processes to transform and analyse data in the frequency domain. For example, plant disease identification was assessed in the frequency domain, where the discrete cosine transform (DCT) converts the images to the frequency domain (Barradas et al., 2022). Another method fused the frequency and spatial domains to develop a dual-branch network (MASF) for crop disease identification (Li et al., 2024). Finally, some methods tried to mitigate the complexity of raw visual inputs indirectly by removing background or segmenting lesion areas on leaves (Liang, 2021; Lin et al., 2022). In addition to a model for FSL, a separate network needs to be trained to prepare inputs for feeding into the FSL network, which causes computational overheads.

Detection of diseases in a reliable and realistic setting remains challenging despite current advances because in-field samples contain not only complex backgrounds but undesired objects. The result of a recent study revealed that the complexity of visual samples directly impacts plant disease detection accuracy (Rezaei et al., 2024a). It is more convenient to achieve satisfactory results when simple visual samples, captured a single disease leaf with a plain background, are accessible (Rezaei et al., 2024a). A few methods are proposed for disease identification in a real-case scenario. For example, Kendler et al. (2022) developed a CNN-based network for crop identification given short-range imaging in fields. However, most of the recent studies neglected this fact. The shortage of in-field datasets is one of the main constraints (Attri et al., 2023). Moreover, recent DL research has overlooked grains and cereals despite being the main human energy sources (Rezaei et al., 2024b). Consequently, in this study, we have captured visual samples of barley disease in the field environment, a realistic scenario. The Barley dataset is refined and altered given the FSL episodic training process in the meta-training step. Moreover, this article presents a method to address data scarcity issues in agriculture, especially for detecting barley diseases. We used the most recent FSL pipeline for barley disease detection (Rezaei et al., 2024a). The method is based on the state-of-the-art FSL method, which does not require additional text inputs within visual samples (Hu et al., 2022). We mainly evaluate the performance of recently developed transformers due to their unprecedented abilities. The pipeline is equipped with a recent type of transformer called the Swin transformer, which is a hierarchical vision transformer using shifted windows. The method is the first to assess the capability of Swin transformers, which achieved high performance only using five labelled images per disease class, as a feature backbone in a few-shot scenario for barley disease detection. To summarise, the main objectives of this study are as follows: (1) to offer a novel DL method based on FSL for barley disease detection, given scarce and complex labelled data, (2) to collect in-field RGB imagery at a trial paddock to construct a new barley disease dataset, (3) to modify, pre-process, and finally repurpose the dataset to meet the episodic training requirements for the K-way N-shot task, and (4) to evaluate the Swin transformer, which is underexplored in agriculture, as the feature extractor backbone.

The rest of the article is organised as follows. In Section 2, the proposed method is described, and the modules in the proposed FSL pipeline are introduced. Data acquisition and preparation are described for barley disease detection. The results of our experiments are presented in Section 3 and discussed in Section 4. The last section concludes the results and findings.

2. Methodology

2.1. A FSL pipeline for barley disease detection

The primary motivation behind the development of FSL is the shortage of data, so it attempts to learn from a few annotated samples given previous knowledge (experienced classes). The support set's size defines the K-way N-shot FSL classification task, meaning that inputs are classified into K classes given N sample per class. Meta-training and meta-testing are the two typical stages of FSL (Vinyals et al., 2016). In addition, the data problem can be alleviated by transfer learning (Ma et al., 2024). As well as few-shot learning, transfer learning increases the reliability of a model facing insufficient training data by transferring the network knowledge across different tasks (Tian et al., 2024). The meta-training step trains the model to classify input data (a query set) in an episodic manner, given the feature vector distribution of a few available samples (a support set). Meta-testing evaluates the model on novel samples (a test set), which the model never sees during the training step. Our model to address the barley disease detection task includes two steps: transfer learning and meta-learning, which are discussed in the following sections.

2.1.1. Transfer learning

Transfer learning, transferring the knowledge achieved on a source domain to a target domain, can attenuate the dataset size problem and enhance performance (Hu et al., 2022). Reusing a backbone pre-trained on a general dataset for a novel dataset has achieved promising results in deep learning to resolve the need for extensive annotated samples, and recently, its capability has been shown in few-shot classification (Liu et al., 2024b; Rezaei et al., 2024a). Pre-training on an auxiliary dataset facilitates extracting meaningful features in the target task. Consequently, the network is trained efficiently with limited training data rather than training a model from scratch. In addition, due to the overfitting problem, training flexible and large architectures, e.g., transformers, may be impractical given a small dataset (Hu et al., 2022). Training millions of network parameters with inadequate input samples generally causes a pre-trained network to outperform the same network without transfer learning. As a result, the backbones of our model benefit from transfer learning because pre-training significantly impacts the performance of FSL tasks where extensive data is unavailable (Rezaei et al., 2024a). Pre-trained networks, trained on the ImageNet dataset (Russakovsky et al., 2015), were employed for barley disease classification in this step. These pre-trained networks, commonly used for various applications, enable the model to use previous knowledge for barley disease detection. Compared to training the DL network from scratch, this approach provides a less complex learning process by adapting a pre-trained network for the target task with fewer samples (Ritharson et al., 2024). In addition, it maintains the model's robustness, assuring reliability and accuracy (Ritharson et al., 2024). The overfitting problem arises when there is insufficient training data because the model may begin to memorise the data rather than generalise (Ma et al., 2024).

Task formulation. Transfer learning is defined as follows. Given a target domain $D_t = \{(x_t^i, y_t^i)\}_{i=1}^{N_t}$, in which x_t denotes a training sample (Ma et al., 2024). The corresponding label is y_t, which can be drawn from a joint data distribution $p_t(\mathbf{x}, \mathbf{y})$. If N_t, which defines the size of the training dataset, is large enough, the training dataset can train a DL model from scratch (Ma et al., 2024). While usually limited data is available, this issue is alleviated with existing large datasets (a source domain). In a classification task, the knowledge gained by a source domain $D_s = \{(x_s^i, y_s^i)\}_{i=1}^{N_s}$, in which x_s, y_s denotes a training sample and the corresponding label in the source domain respectively given a joint data distribution $p_s(\mathbf{x}, \mathbf{y})$. It is assumed that N_s is quite large (Ma et al., 2024). Nevertheless, the performance of DL models may be significantly reduced if a pre-trained network is directly applied to the target domain without tuning due to domain shift, which is the variations in the target and source domains' data distributions ($p_t(\mathbf{x}, \mathbf{y}) \neq p_s(\mathbf{x}, \mathbf{y})$) (Ma et al., 2024). The aim is to leverage the achieved knowledge from the source domain to improve performance in the target domain (Ma et al., 2024). The pre-trained network has learned to extract useful features, including edges, lines, and textures, which can easily transferred across different tasks. This general knowledge is beneficial for other tasks (Ritharson et al., 2024). This technique exerts knowledge transition from a source domain, which has abundant data, to a similar target domain that suffers from a lack of sufficient data.

2.1.2. Meta-training

Meta-training is the procedure of training the network to learn from available meta-knowledge, or as reported, "learn to learn", and it has numerous applications, including architecture search, reinforcement learning, and computer vision (Tian et al., 2024). Meta-training quickly adapts the model and generalises its classification capabilities to novel classes with limited samples by simulating an episodic training approach (Liu et al., 2024b). The process uses the extracted experience through previous learning episodes to train the model for future learning tasks. It adopts the knowledge gained from various learning experiences to enhance performance in the new tasks (Tian

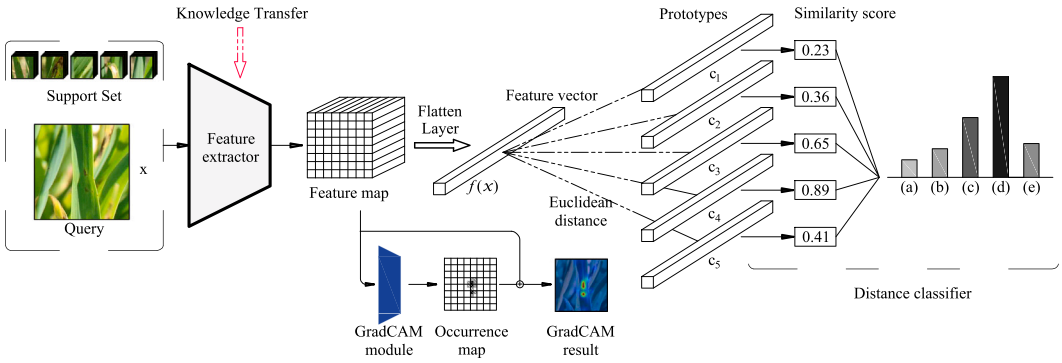


Fig. 1. A visualisation of the method's structure, showing an input task including a support set and a query sample x, the backbone extracting the feature map, GradCAM module calculating occurrence map, classifier layer classifying based on Euclidean distance between a feature vector and prototypes, and output classes (a-e).

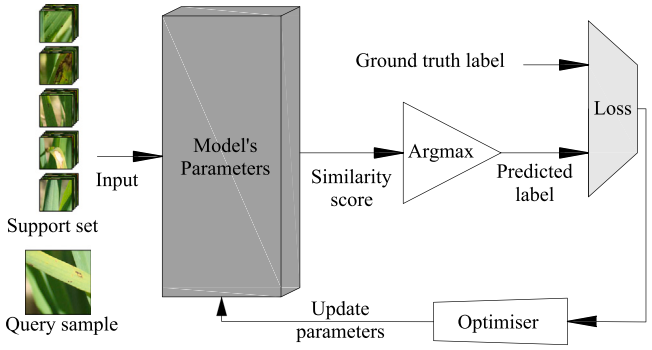


Fig. 2. A visualisation representation of the meta-training, given the input comprising a support set and a query image.

et al., 2024). Meta-learning is persistent with FSL since limited samples are available for future tasks, in which a model is trained on a small number of new samples alongside previous experiences. Consequently, this training process has been widely employed in FSL to facilitate the learning process (Tian et al., 2024). The method gains generalisation ability by training the model on several source tasks, also known as training tasks, divided into support and query sets (Tian et al., 2024). Meta-learning trains the model to classify query inputs given corresponding tasks, a training sample set, rather than a single sample. Tasks consist of a query set for evaluation and a support set for training. A single backbone, for example, Swin or ViT, extracts features of input samples, and the model uses the prototypical classifier (Chen et al., 2018) to measure similarity scores between the support set and a query sample. By calculating the mean vector of the support set embedding vector extracted by the backbone, the model generates the class centroids, prototypes, in each episode. Finally, the input is classified based on its similarity to prototypes.

Task formulation. A few-shot task has only a few samples for training. The K-way N-shot classification is a K-class classification in which only N samples per class (a total number of $K \times N$) are available for training. Generally, this problem can be defined as a various-way-various-shot classification. In contrast to training a classic data-driven network, the meta-training procedure is task-driven because support and query sets generate a task provided to the FSL network to train parameters. The training samples are divided into different tasks T_i. Each training task is split into a query set Q and a support set S as defined in Eqs. (1) and (2). The query set Q comprises N_q images per class, so $K \times N_q$ images are available for testing the FSL performance. As defined in Eq. (1), the support set totally has $K \times N$ labelled sample, i.e., N samples per class. $x_k^n \in \mathbb{R}^D$ is the D-dimensional vector corresponding sample n of class k, and its label is denoted by y_k.

$$S = \{(x_k^n, y_k)\}_{n=1}^{N}, \quad \forall\, k \in \{1, \ldots, K\} \tag{1}$$

$$Q = \{(x_k^q, y_k)\}_{q=1}^{N_q}, \quad \forall\, k \in \{1, \ldots, K\} \tag{2}$$

As defined in Eq. (3), prototypes, $c_k \in \mathbb{R}^M$, are the mean vectors of the embedding feature vectors of the support set, which are extracted by the backbone function $f_\phi : \mathbb{R}^D \to \mathbb{R}^M$. As a result, the mean vector of all samples of a class k is computed as the classes' prototypes in Eq. (3). Trainable parameters of the backbone embedding function f_ϕ are specified as ϕ.

$$c_k = \frac{1}{K} \sum_{n=1}^{N} f_\phi(x_k^n) \tag{3}$$

The softmax generates a categorical distribution over the query sample $\mathbf{x} \in Q$ based on the similarity scores between class prototypes and input embedding features obtained by a distance function d. Eq. (4) provides the probability that a query sample $\mathbf{x}$ is assigned to a class k.

$$p_\phi(\mathbf{y} = k|\mathbf{x}) = \frac{\exp(-d(f_\phi(\mathbf{x}), c_k))}{\sum_{\acute{k}=1}^{K} \exp(-d(f_{\phi(\mathbf{x})}, c_{\acute{k}}))} \tag{4}$$

In meta-training, the cross-entropy loss, Eq. (5), is minimised by the stochastic gradient descent (SGD) optimiser.

$$L_{CE}(x, y) = -\sum_{i=o}^{K} y_i \log(p_{\phi_i}) \quad for\ K\ classes \tag{5}$$

Pseudo-code in Algorithm 1 summarises the details of the training process, and the model's structure is depicted in Fig. 1. The model comprises a backbone for feature extraction and a distance classifier measuring the similarity of an input query and class prototypes. The backbone encodes an input to the feature map and then, after the flatten layer, the embedding feature vector. An input is classified based on the Euclidean distance between the extracted feature vector and the prototype of each class. The overview of the meta-training step is presented in Fig. 2. In each meta-training episode, task-level input data, a task T_i, has a support and query set. The classifier trains to determine the class of each query image given the most similar prototype. The optimiser updates the model's parameters using the cross-entropy loss. ResNet50, ViT, Swin-T, and Swin-B are used as the model's backbone, extracting the class-related feature vectors. For Swin transformers, to reduce the output feature map resolution and thereby lower the computational cost, a pooling layer was added that calculates the average pooling of the last hidden state layer.

2.2. The imaging system

In this data acquisition task, the imaging system is a 24.1 megapixel Pentamirror CANON EOS 2000D DSLR camera, with the EFS 18–55 mm MACRO $0.25m/0.8ft$ lens. The RGB camera captures raw images with a resolution of 4000 × 6000 pixels, 72 dpi, and 24 bit. The lens focal length was configured to 28 mm, and the other settings were fixed as

Algorithm 1: Pseudo code for barley disease detection using FSL, the meta-learning procedure.

Data: $batch \leftarrow S, \mathbf{x} \in Q$
`# x is a query sample`
Result: $\mathbf{y} \in \{1, \cdots, K\}$
`# y is a disease class`
Start
Step 1**:** Transfer learning
$f_\phi \leftarrow backbone$ `# Init`
Step 2**:** Meta-training
while $i \leq epoch$ **do**
  $stats \leftarrow train(batch, f_\phi)$;
  while $j \leq batch$ **do**
    $S, x, y \leftarrow batch$;
    $P_k \leftarrow mean(f_\phi(S))$ `# for each class`
    $outputs \leftarrow f_\phi(x)$;
    $logits \leftarrow d(P_k, outputs)$;
    `# Euclidean distance`
    $loss \leftarrow CrossEntropy(logits, y)$;
    $optimize(loss.backward())$;
    $loss_{val} \leftarrow test(Q_{val}, f_\phi)$;
    `# evaluation accuracy`
    if $loss_{val} \leq loss_{val}$ **then**
      save model
    end
  end
end

follows: $EXP = 0.7$ and $ISO = 100$. The camera automatically adjusts the shutter speed according to the target's brightness. The camera is configured with a low aperture $f/4.5$ to create a slightly blurred background. We manually triggered the camera that was positioned at a high angle and a distance of approximately one meter from the target crop.

2.3. Data acquisition and dataset description

In this research article, a dataset of barley disease is constructed by capturing visual samples of diseased and healthy crops in the barley field. Because complex samples contain undesired objects, making them more complicated to work, this research follows a real case scenario, in which images are acquired in the field with complex backgrounds and environmental conditions, rather than a lab experiment. Due to its importance, barley is selected as a target of this research project. To diversify the samples, we captured 312 RGB images of the barley leaves in natural daylight under various weather circumstances, such as cloudy and sunny. Visual samples of barley leaves infected with the three common diseases are collected at a barley trial paddock located at the Department of Primary Industries and Regional Development (DPIRD), South Perth, Western Australia, Australia. Data was taken at different growth stages of barley disease, $6-8$ weeks after planting and $3-5$ weeks after disease inoculation. The total number of three identical barley trials were inoculated with NFNB, SFNB, and scald. In these trials, both plot length and spacing between plots are 0.5 m. The expert (the fourth author) supervised data collection and annotation tasks to minimise possible errors.

The dataset samples are generated by cropping raw input images into uniform patches of 448 × 448 pixels. Through this patching process, the number of samples is expanded to a total of 2 165. To establish the ground truth data, the extracted patches are manually annotated under the guidance of experts into five classes: no-disease, scald, early NFNB, severe NFNB, and SFNB. Fig. 3 shows samples of the Barley dataset, in which different disease symptoms can be observed. Symptoms of scald disease appear as grey-green lesions with brown margins, in which shapes can be irregular. Dark-brown spots (elliptical lesions) are visible symptoms of SFNB. For NFNB, the infection stages vary from early to severe. So, this dataset creates two classes for NFNB. In the early stage of NFNB, dark-brown lesions are elongated in a net-like form. These lesions gradually expand to the whole surface of infected leaves in the severe stage of NFNB. Fig. 3 shows a raw sample captured from the field, and it represents the complexity of samples in our dataset to visually illustrate the difficulty of the task. As shown in Fig. 3, samples contain complex backgrounds. Table 1 shows the number of available samples for each class. The Barley dataset contains 246 samples of early NFNB, 156 samples of severe NFNB, 280 samples of SFNB, 175 samples of scald, and 1 308 samples with no-disease. Fig. 4 presents the distribution of samples across the Barley dataset. The practicality of our Barley dataset was previously examined in a typical image classification task (Rezaei et al., 2024b).

Table 1
Barley dataset summary.

Classes	Samples
No-disease	1308
Spot form net blotch (SFNB)	280
Scald	175
Net form net blotch (NFNB)	246
Severe NFNB	156
Total	2165

2.4. Data selection

Based on FSL requirements, the dataset was regenerated to adopt our few-shot setting. The Barley dataset is randomly divided into three sets: the training set S_{train}, the validation set S_{val}, and the test set S_{test}, with a ratio of 65%, 15%, and 20% respectively. Evaluation and testing samples are never seen during the training step. Training, validation and test sets are disjoint ($S_{train} \cap S_{test} \cap S_{val} = \emptyset$), and validation and test samples are never seen during training. The model performs episodic training, in which training consists of several tasks T_i that is a subset of training samples consisting of a support set, a small number of samples, and a query set. Samples are selected randomly, and corresponding labels are one-hot encoded in each support and query set. The same process is applied to the test and validation set to generate evaluation tasks. Here, the FSL problem is the 5-way 5-shot classification task. This five-class classification exploits features of only five samples per class (total of 25 samples), available in each support set, for training. Finally, the model classifies query samples based on their similarity to support samples. To create a consistent plant disease dataset for the few-shot task. 25 random samples (5 samples per class) create the support set, and 15 random samples create the query set in each task.

2.5. Data pre-processing

The annotated samples are prepared for feature extraction with image pre-processing methods. The pre-processing step comprises some random adjustments (colour jitter) for training purposes in this study. Adding random input noise is an approach to improve generalisation in deep neural networks (Jafarigol and Trafalis, 2023). First, all samples are resized to 224 × 224 pixels as the size of images is 448 × 448 pixels. Moreover, the colour jitter transform is exerted on inputs to pre-process inputs in the training step. This transform allows us to randomly adjust the brightness, contrast, and saturation with the following probabilities: $brightness = 0.1$, $contrast = 0.1$, and $saturation = 0.1$. The parameter was selected among two values (0.1 and 0.4) through experimental evaluation. Then, we performed random horizontal flips (with a probability of 0.5) and image normalisation. These random functions are applied to enhance the model's generalisability in the training procedure. To evaluate the method on actual samples, test samples are normalised without adding a random transform.



(a) Raw images (b) Output samples of classes

Fig. 3. Visual samples of the Barley dataset, showing (a) a raw image captured from the barley trial paddock, and (b) output samples of each class, which are generated by cropping and annotating the raw images.

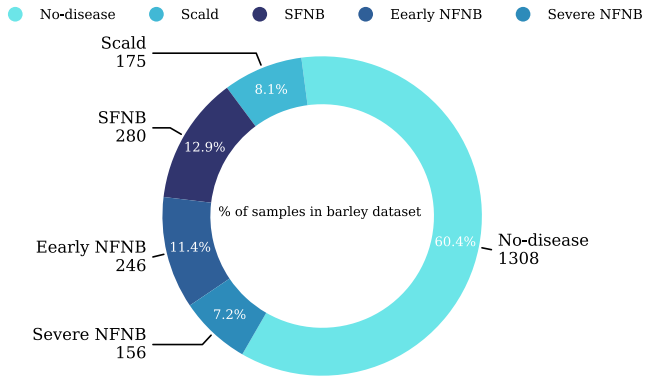


Fig. 4. Distribution of samples across the Barley dataset.

2.6. Supplementary dataset characteristics

A limited number of plant disease detection datasets are available online, especially those captured in field trials (Attri et al., 2023). We used a dataset, which was manually collected in-field leaf images using a commercial camera, to assess our method. Like our Barley dataset, the cassava dataset includes varied field backgrounds and undesired objects (Ramcharan et al., 2017). The dataset was regenerated and repurposed to be applicable to our task. First, five classes were randomly chosen. Second, raw samples would have been discarded if they included unexpected objects, such as the photographer's hand. Finally, since the resolution of raw cassava samples is higher than our dataset samples, through a pre-processing step, samples were cropped into smaller patches to cover only the leaf area in the centre of samples, as shown in Fig. 5. The cassava dataset's final version comprises 636 samples in five classes. The class name and corresponding sample number are as follows: 115 samples of brown leaf spot (BLS), 129 images of mosaic disease (MD), 120 samples of green mite damage (GMD), 121 samples of red mite damage (RMD), and 151 samples of the healthy class. Fig. 5 displays examples of each class in the regenerated cassava dataset.

2.7. Parameter setting

To evaluate our 5-way 5-shot scenario, the number of classes is 5 in the training and testing episodes, and there are 5 images in each support set class and 15 images in each class of the query set. In the 5-way 1-shot scenario, only one image is available for each class. We considered 2 000 episodes for training and testing, and the number of episodes for validation was 1 000. The model was trained for 50 epochs using the stochastic gradient descent (SGD) optimiser. As recommended for few-shot settings, the momentum factor was set to 0.9, and a warm-up plus cosine annealing learning rate was used to reduce the overfitting risk (Hu et al., 2022). The learning rate, which its initial value is 1×10^{-5}, increases to 5×10^{-5} after five epochs and then decreases to its initial value. A recommended setting was followed in the FSL task to select the values (Rezaei et al., 2024a; Hu et al., 2022). Cosine and Euclidean similarities are commonly used in FSL. However, the Euclidean similarity was chosen since it achieved slightly better results on the Barley dataset. Default values are used for backbone configuration.

2.8. Test environment

The following hardware was used to test the proposed method on the Ubuntu 20.04.6 LTS Operating System: CPU: Intel(R) Core(TM) i9 – 10900K CPU @ 3.70 GHz, Graphic card: NVIDIA GeForce RTX 3060 12 GB CUDA Version 11.4, and Memory: 32G × 3 DDR4 Synchronous 2933 MHz (0.3 *ns*).



Fig. 5. Visual samples of the cassava dataset (Ramcharan et al., 2017), showing five classes: (a) brown leaf spot (BLS), (b) mosaic disease (MD), (c) green mite damage (GMD), (d) red mite damage (RMD), and (e) healthy.

3. Results

This section provides results to discuss the classification performance of FSL on barely disease detection. Our results demonstrate the FSL performance on the captured barley dataset. First, to evaluate the impact of data acquisition, we captured samples, including complex backgrounds and undesired objects, in the barley field. The barley disease dataset is prepared to evaluate the FSL method in barley disease detection. In a realistic scenario, it has been assumed only a limited number of samples are accessible in FSL. The efficacy of our method is shown by conducting extensive evaluations on the Barley disease dataset, and this case experimentally approves the efficiency of FSL. Here, the model benefits from transfer learning because the impact of transfer learning is crucial in FSL task (Rezaei et al., 2024a).

Second, as Table 2 shows, we experimentally evaluated the fine-tuned model (without meta-training) on the Barley dataset to highlight the impact of meta-training, because it has been reported recently that achieving promising results may result from a pre-trained network, and the advanced episodic training impact of meta-training has been suspected to be negligible (Liu et al., 2024b).

Third, the results show mainly meta-training performance as pre-trained networks are used in our model. The impact of meta-training in FSL is highlighted. Fourth, the network's performance in recognising barley disease is evaluated with multiple backbones. The network's backbone is equipped with several vision transformers, which are meta-trained on the Barley dataset, to assess barley disease detection capability. In this setting, different transformers are deployed to extract beneficial features of each class. Two base transformers (Swin-B and ViT) are compared to evaluate the transformer's ability to extract desired features in barley samples. In addition, Backbones with less trainable parameters (ResNet50 and Swin-T) are used in this study to make a practicable method for barley disease detection.

Finally, the ability of each backbone is further assessed on the Barley dataset for each barley disease in a few-shot scenario. The detection accuracy of test sets is measured after meta-training on the Barley dataset. We compared transformers as the backbone of our model in the results. Table 3 summarises the meta-training performance of the FSL pipeline for barley disease detection. As shown in Table 3, Swin-B achieves the highest detection accuracy for 5-way 1-shot and 5-way 5-shot with 86.78% and 91.88% accuracy, respectively. One shot means only one training image per class was used. To validate the method's practicality, the method's performance is similarly evaluated on the cassava dataset (Ramcharan et al., 2017), which is shown in Table 4.

The results indicate that the FSL model robustly works on the Barley dataset with less training data. Transformers are great feature extractors despite their complexity, especially for complex samples. The feature extraction ability of each backbone is investigated by extracting the class occurrence map using gradient-weighted class activation mapping. The examples are visualised in Fig. 6, which shows how each backbone localises class-related features. For further investigation, we measured the effectiveness of different backbones for each class of the Barley dataset. The confusion matrix in Fig. 7, shows the detection accuracy for each class separately. For example, as shown in Fig. 7, all transformers correctly classify no-disease samples.

Table 2
Classification accuracy of the fine-tuned model (without meta-training) for one-shot, five-shot, and ten-shot tasks, comparing backbone feature extractors.

Backbone	5-way 1-shot	5-way 5-shot	5-way 10-shot
ResNet50	35.55	60.77	72.49
Swin-T	36.48	66.02	75.95
ViT-B	37.50	63.64	74.93
Swin-B	**38.63**	**66.46**	**78.15**

Table 3
Classification accuracy after meta-training for one-shot and five-shot tasks, comparing backbone feature extractors.

Backbone	Accuracy		N param
	5-way 1-shot	5-way 5-shot	
ResNet50	83.10	87.36	23M
Swin-T	83.29	90.45	27M
ViT-B	84.91	90.88	85M
Swin-B	**86.78**	**91.80**	86M

Table 4
Classification accuracy on the cassava dataset for one-shot and five-shot tasks, comparing backbone feature extractors.

Backbone	Accuracy	
	5-way 1-shot	5-way 5-shot
ResNet50	92.11	89.63
Swin-T	93.87	91.61
ViT-B	96.13	91.50
Swin-B	**97.93**	**95.24**

4. Discussions

The following sections discuss the results, which are evaluated on the captured Barley dataset and another supplementary dataset. First, the ability of Swin transformer to extract features of complex images is investigated for the defined FSL task in agriculture. Second, given the Barley dataset, the FSL model is evaluated in two 5-way 1-shot and 5-way 5-shot settings. In addition, the model is evaluated on a supplementary dataset. Lastly, a comprehensive investigation of misclassification items is conducted to complete the results.

To determine the classifier's ability to detect barley disease, The detection accuracy of different backbones is shown in Table 3. Comparing feature extraction ability, transformers outperform ResNet50, which is a classic DL feature extractor. Among transformers, Swin-B archives the highest detection accuracy.

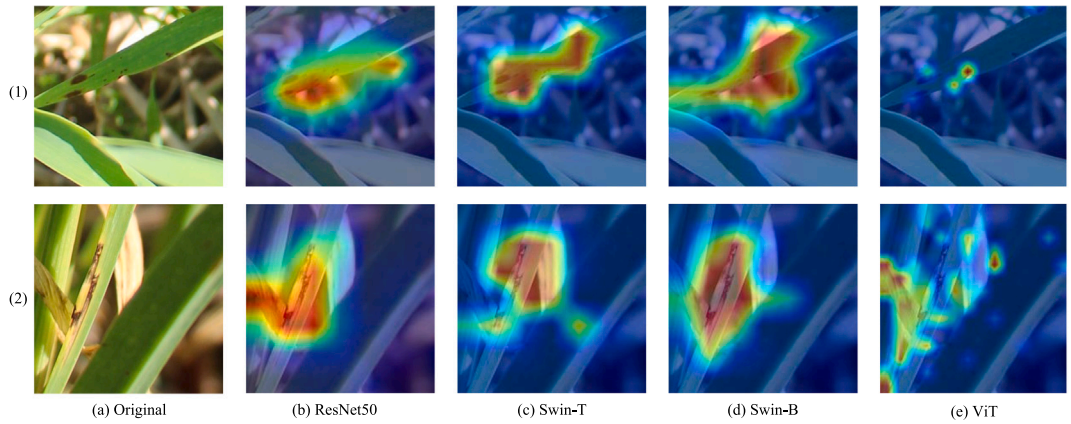


Fig. 6. Exemplary visualisation of GradCAM results, implemented on (b) ResNet50, (c) Swin-T, (d) Swin-B, and (e) ViT backbones. Only ViT misclassify disease type in these examples. Major disease parts are correctly highlighted.

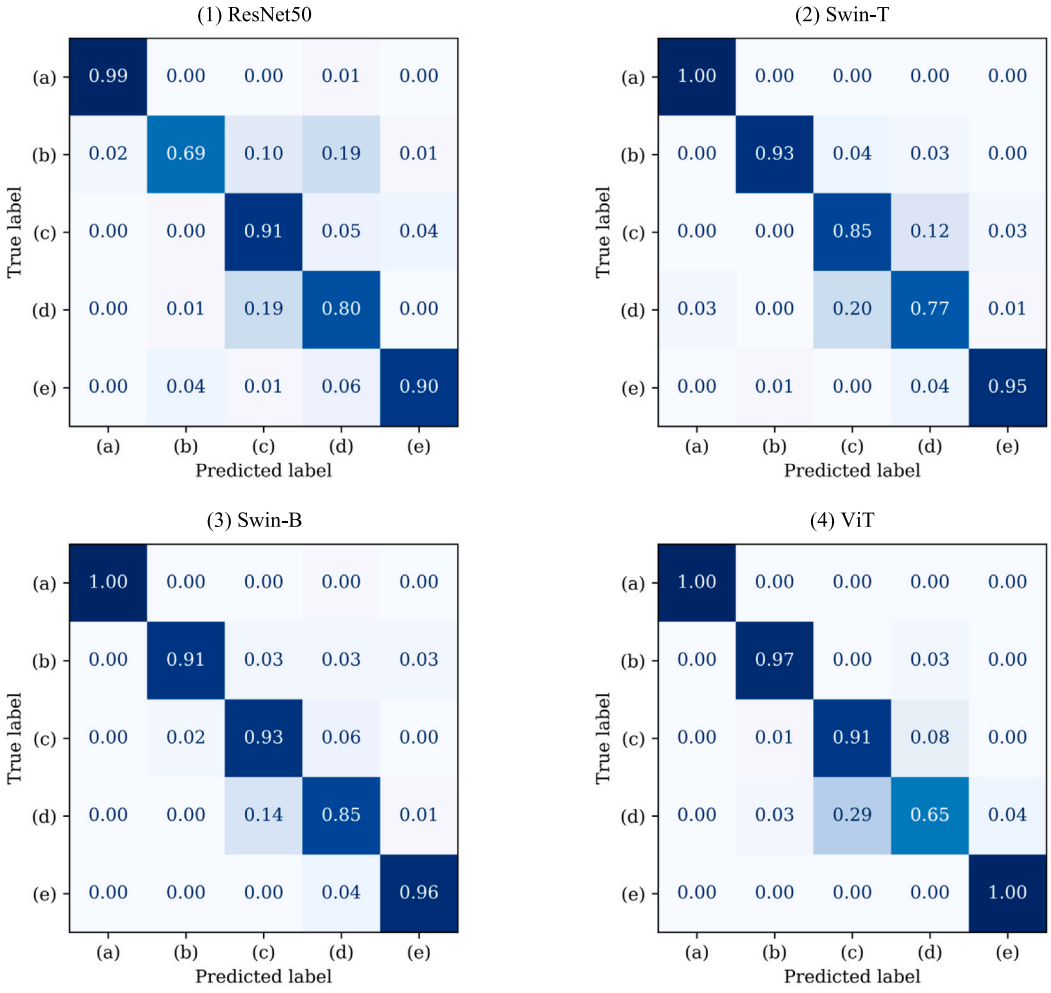


Fig. 7. Confusion matrix for (1) ResNet50, (2) Swin-T, (3) Swin-B, and (4) ViT. In each figure, labels are defined as follows: (a) no-disease, (b) scald, (c) SFNB, (d) early NFNB, and (e) severe NFNB.

4.1. Meta-training impact

It was noted that the meta-training impact might be minimal, and the high performance may be achieved due to transfer learning (Liu et al., 2024b). As a result, an experiment is implemented to evaluate and highlight the performance of the meta-training step. First, the fine-tuned model, which utilises fine-tuning on the transferred model without meta-training, is evaluated on the Barley dataset in Table 2. On the other hand, Table 3 shows the evaluation result after meta-training to underscore the effect of this step. As shown in Table 2, without meta-training, Swin-B achieved the maximum detection accuracy of 38.63% and 66.46% in the one-shot and five-shot tasks, respectively. After meta-training, the model performance is improved, and Swin-B gained the detection accuracy of 86.78% in the one-shot task and 91.80% in

the five-shot task, showing both transfer learning and meta-training have a noticeable influence on performance.

4.2. Swin transformer in disease detection

Swin transformers are versions of vision transformers, which compute a hierarchical feature representation with shifted windows (Liu et al., 2021). These hierarchical transformers are based on the shifted windowing process, which is shown to be an efficient self-attention mechanism (Liu et al., 2021). As experimented on computer vision datasets, it was claimed that the Swin transformer is a general-purpose feature extractor, that is capable of solving different tasks (e.g., image classification, object detection, and semantic segmentation) in computer vision (Liu et al., 2021). Although the size of the Swin model is larger than ViT, its theoretical computational complexity is slightly lower (Liu et al., 2021). ViT has quadratic computational complexity to sample size because self-attention is computed globally (Liu et al., 2021). On the other hand, the Swin transformer has linear computational complexity to sample size since it computes self-attention only inside each local window and creates feature maps by combining the sample's patches in deeper layers (Liu et al., 2021). With similar throughput, the Swin transformer outperformed ViT in image classification on the ImageNet dataset, demonstrating its strong feature extraction capability as a backbone (Liu et al., 2021). This novel transformer is underexplored for small datasets. In addition, the capability of Swin transformers is little-known in agriculture, especially for FSL. Consequently, the performance of the Swin transformer in FSL is investigated in this project. Transformers extract class features by dividing images into patches that are inputs of a transformer-like architecture. This process allows transformers to learn an internal representation of the input. In contrast to ViT, which uses fixed-size windows, Swin transformers exert variable-size windows and gain a deeper understanding of local features (Liu et al., 2021). Transformers outperform classic ResNet50 for complex samples or other objects, because they use a sequence of patches, as shown in Table 3. In addition, comparing transformers, the base form of Swin transformers (Swin-B) achieves the best result on the Barley dataset in 5-way 1-shot and 5-way 5-shot tasks.

4.3. Gradient-weighted class activation mapping (Grad-CAM)

DL models are considered black boxes as they cannot provide interpretable outputs. Understanding what happens inside the network and why it fails to achieve satisfactory results is essential. GradCAM offers graphical representations of model predictions by localising the target object and providing the object occurrence map (Selvaraju et al., 2016). GradCAM's output is a heatmap visualisation for a target class. The last layer's average gradient is used to weight the 2D class occurrence map (Selvaraju et al., 2016). It highlights the class-related regions by projecting the weights of the last layer on the feature occurrence maps, which mark the parts that significantly contributed to classifying the target. GradCam assists us in understanding how a DL network makes a decision. We monitored the gradient information of the last layer before the distance classifier.

Fig. 6 presents the results of the GradCAM module after meta-training. As the input images contain undesired objects and field backgrounds, some parts may contain more related information about the classes. As shown in Fig. 6, all backbones highlighted the class-related parts of samples, such as symptoms. Although the ViT achieves satisfactory results, Fig. 6 shows two examples that ViT misclassified, but the other backbones correctly classified samples. In general, the output of GradCAM heatmaps shows that our model can locate the disease region and identify the parts where apparent symptoms occur, highlighting our model's localisation ability, which can be further used in automatic pixel-wise annotation in future studies.

4.4. Supplementary dataset analysis

To further evaluate the generalisability of our FSL approach to other crops and validate the method's effectiveness, the method's performance is assessed on the cassava dataset (Ramcharan et al., 2017). The dataset is relatively similar to our Barley dataset since both manually captured in-field samples containing complex backgrounds and unwanted objects. The same methods were applied to the cassava dataset for a fair comparison. Table 4 shows the achieved detection accuracy on the cassava dataset. Comparing the feature extraction backbones, the method achieved the same results. In other words, Swin-B outperformed other backbones on the cassava dataset and achieved 97.93% and 95.24% detection accuracy in 5-way 5-shot and 5-way 1-shot tasks, respectively. On the other hand, ResNet50 achieved the worst detection accuracy of 92.11% and 89.63% in 5-way 5-shot and 5-way 1-shot tasks, respectively. Overall, the methods achieved better detection accuracy on the cassava dataset since classifying barley disease is more challenging due to the shape of barley leaves.

4.5. Prediction summary

Fig. 7 represents the confusion matrix (or error matrix), summarising the detection ability. Fig. 7 aids in assessing the performance of our FSL model as it displays the number of correct and incorrect predictions and identifies misclassifications, which are separated for each class. As misclassifications may happen in one class only, classification accuracy cannot highlight that, leading to misleading interpretations. Consequently, the confusion matrix is calculated in Fig. 7, which gives us insight into errors and understanding of the model's confusion. One fixed series of samples, randomly chosen from the test set, is fed to the model in this experiment to organise the order of test samples to be the same for all backbones.

As shown in Fig. 7, major misclassifications of the ResNet50 backbone occur in the scald class; only 69% of samples are accurately classified, and the backbone misclassified 19% of this class to early NFNB. In addition, ResNet50 made errors in 20% of early NFNB samples, which are mostly confused with the SFNB class. On the other hand, transformers correctly classified more than 90% of samples in the scald class. However, SFNB and early NFNB are two classes that are being confused by transformers (Swin-T, Swin-B, and ViT). Most misclassifications happen in SFNB and early NFNB classes, in which the symptoms are not easy to distinguish in the early stage. In differentiating between these two classes, Swin-B achieved the best performance.

5. Conclusions

Many DL methods have been proposed to recognise plant disease; however, classic DL cannot succeed with inadequate data because the training dataset size substantially impacts its performance. Moreover, collecting data is laborious and may not be possible in some agricultural cases. Consequently, this method is designed for small training datasets, which is the most common case, to mitigate the model's reliability on the size of datasets. In addition, this study aims to add another layer of difficulty to plant disease detection and evaluate the model in a real-case scenario, in which data includes samples with complicated backgrounds and unwanted objects. A novel barley dataset is prepared to achieve the objective. Barley disease samples are collected at a trial paddock, pre-processed, and finally labelled into five classes: NFNB, SFNB, scald, severe NFNB, and no-disease. This article utilised the most recent FSL method for barley disease detection given limited visual samples (five samples per class). We investigated the impact of transformers as a feature backbone in FSL for barley disease detection. Moreover, we further examined the method on the cassava dataset and verified how well transformers perform in plant disease detection. In few-shot classification, our approach is the first to qualitatively evaluate the essence of the Swin transformer for

plant disease recognition. The main findings can be summarised as follows. FSL pipeline equipped with transformers effectively achieves satisfactory detection accuracy with limited data, especially when it was equipped with Swin transformers. Comparing transformers as the feature extractor, the Swin transformer outperformed ViT, achieving 91.80% and 97.93% detection accuracy on barley and cassava datasets, respectively. Since the result is data-driven, this does not prove that the Swin transformer is superior to ViT in different datasets. Meta-training substantially impacts the performance of models, making it a necessary element where data is Given the misclassification results, the backbones confused samples of SFNB and early NFNB due to their visual similarities; however, Swin-B distinguished these classes better. This advanced AI method is capable of distinguishing objects by only a few examples in 5-way 5-shot and 5-way 1-shot tasks. Given visual GradCAM outputs, the model is able to localise symptoms and ignore unrelated information, such as backgrounds.

This paper contributes towards advancing machine learning approaches that can work well in data-scarce contexts. One of the limitations of the work is the complexity of transformers. An advanced system configuration is necessary to train these heavy networks. However, the method can further be simplified using feature distillation methods in future studies. The GradCAM results indicate that the feature backbone can highlight disease symptoms, which can be used for lesion segmentation in future studies. Another limitation is manual data annotation, which requires significant time and effort to generate the dataset. Future research can expand the task to open-world few-shot learning (OFSL), which incorporates unreliable labels within scarce samples.

CRediT authorship contribution statement

Masoud Rezaei: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dean Diepeveen:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Hamid Laga:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Sanjiv Gupta:** Writing – review & editing, Writing – original draft, Resources, Data curation. **Michael G.K. Jones:** Writing – review & editing, Supervision, Resources. **Ferdous Sohel:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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